Learning Objective 9.1: I understand the process of decoupling and finding solutions for Maxwell's equations for dynamic fields.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **A wave in a lossless dielectric.** A 1 GHz wave travels in the $+z$ direction in a lossless dielectric with $\epsilon_r = 9$ and $\mu_r = 1$. Its electric field is along $\hat{\mathbf{x}}$, with amplitude 10 V/m and phase zero at $z = 0$.

   (a) Find β, λ, and v_p.

   $\beta =$ ____

   $\lambda =$ ____

   $v_p =$ ____

   (b) Write the phasor $\widetilde{\mathbf{E}}(z)$ and the time-domain field $\mathbf{E}(z,t)$.

2. **The exact attenuation constant.**

(a) Starting from $\gamma^2 = j\omega\mu\,(\sigma + j\omega\epsilon)$ with $\gamma = \alpha + j\beta$, derive the two equations $\alpha^2 - \beta^2 = -\omega^2\mu\epsilon$ and $2\alpha\beta = \omega\mu\sigma$, and from them the exact expression for α.

(b) Wet earth has $\sigma = 0.01$ S/m and $\epsilon_r = 10$. Find $\sigma/\omega\epsilon$ at 1 MHz and at 1 GHz, and say which limit, if either, applies at each.

(c) Find α at each frequency from the matching limit, and compare with the exact values, 0.193 Np/m at 1 MHz and 0.596 Np/m at 1 GHz.

α, 1 MHz =

α, 1 GHz =

3. **Skin depth.** Copper has $\sigma = 5.8 \times 10^7$ S/m and aluminum 3.7×10^7 S/m; both have $\mu = \mu_0$.

   (a) Find the skin depth of copper at 60 Hz, 1 MHz, and 10 GHz.

   60 Hz = ____

   1 MHz = ____

   10 GHz = ____

   (b) A shield of aluminum foil $16\ \mu$m thick surrounds a circuit. At 100 MHz, what fraction of the field amplitude that enters the foil reaches its far side? Express the reduction in dB.

   fraction = ____

   reduction = ____

   (c) Seawater has $\sigma = 4$ S/m and $\epsilon_r = 81$. At 10 kHz, where it is a good conductor, find the depth at which the field amplitude falls to 1% of its value at the surface.

   depth = ____

Documentation: _____